3.8 Functions from Their Addition Laws

WORKSHEET

The question

STEP III 2019, Question 2

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The equation holds for *all* x and y. What is the most useful thing to try first?

STAGE 1 The value at zero

Ask yourself What goes wrong if $f(0) = 0$?

STAGE 2 The derivative everywhere from the derivative at zero

Ask yourself In $(*)$, write $f(x+h) = f(x)f(h)$. What can you take outside the limit?

STAGE 4 The second rule

Ask yourself Put $y = 0$ in the rule for g. What does that tell you about $g(0)$? Where does $|g(x)| < 1$ come in?

Pause and try Now compute $g(x + h) - g(x)$ using the rule, simplify, divide by h and let $h \to 0$.

STAGE 5 Solving it

Ask yourself Separate the variables. Which function has derivative $\frac{1}{1-g^2}$ for $|g| < 1$?

What's really going on

Velocities in relativity

In everyday life, if a train moves at speed u and you walk forwards inside it at speed v, your speed relative to the ground is $u + v$. Einstein showed in 1905 that this is only an approximation. The correct rule, with c the speed of light, is

$$u \oplus v = \frac{u + v}{1 + uv/c^2}.$$

Measuring speeds as fractions of c, this is exactly the rule for g. Our result says that if we write each velocity as $v = \tanh\phi$, then velocities combine by *adding* the ϕs. The quantity $\phi = \operatorname{artanh} v$ is called the *rapidity*. Each boost from a rocket engine adds a fixed amount of rapidity, but since $\tanh\phi < 1$ for every ϕ, the velocity creeps closer and closer to the speed of light without ever reaching it (Figure 3.8.1). Combining $0.9c$ with $0.9c$ gives not $1.8c$ but $\frac{1.8}{1.81}c \approx 0.9945c$.

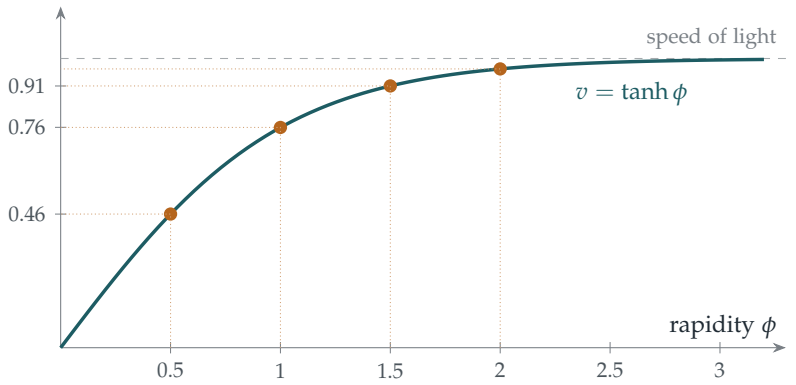


Figure 3.8.1 Velocity (as a fraction of the speed of light) against rapidity. Equal steps in rapidity, which is what equal boosts from an engine provide, give smaller and smaller increases in velocity: 0.46, 0.76, 0.91, 0.96, ..., never reaching 1.

Cauchy's functional equation

The simplest functional equation is $f(x + y) = f(x) + f(y)$, studied by Cauchy in 1821. If f is differentiable, the method of this question shows $f(x) = kx$. In fact continuity is enough. But without any regularity assumption at all, there are other, utterly wild solutions, whose graphs are scattered densely over the whole plane. Georg Hamel constructed them in 1905; the construction needs the *axiom of choice*, and it is impossible to write any of them down explicitly. This is one reason the question insists that f is differentiable: the hypothesis is not a formality.

Homomorphisms

$f(x+y) = f(x)f(y)$ says that f turns the addition of real numbers into the multiplication of positive real numbers, respecting the structure of both. In the language of first-year algebra, f is a *group homomorphism* from $(\mathbb{R}, +)$ to $(\mathbb{R}_{>0}, \times)$, and the logarithm is its inverse. Many of the most important functions in mathematics are characterised in this way, by the structure they preserve. In Section 7.6 we meet the geometric meaning of tanh: it measures "hyperbolic angle", and its addition law comes from composing hyperbolic rotations.

Try this next

1. **A tangent rule.** Suppose g is differentiable, $g'(0) = k \neq 0$, and $g(x+y) = \dfrac{g(x)+g(y)}{1-g(x)g(y)}$ whenever both sides are defined. Show that $g'(x) = k(1+g(x)^2)$ and deduce that $g(x) = \tan kx$ near $x = 0$.

2. **Cauchy's equation.** If f is differentiable and $f(x+y) = f(x)+f(y)$ for all x, y, show that $f(x) = kx$.

3. **Without derivatives.** HARDERSuppose only that $f(x+y) = f(x)f(y)$ and $f(1) = a > 0$. Show that $f(n) = a^n$ for positive integers n, then that $f(1/n) = a^{1/n}$, and then that $f(q) = a^q$ for every positive rational q. (Why is continuity needed to go further?)

4. **Rocket science.** A rocket can increase its speed by $0.5c$ relative to its current frame each time it fires. Starting from rest, what is its speed (relative to the start) after 2, 3 and 4 firings? Use rapidities.

Further reading

- **Christopher Small, *Functional Equations and How to Solve Them*** (Springer). An accessible introduction aimed at problem-solvers.
- **Edwin Taylor & John Wheeler, *Spacetime Physics*** (W. H. Freeman). The classic introduction to special relativity, using rapidity throughout.
- **Tom Körner, *Calculus for the Ambitious*** (Cambridge University Press). Defines the exponential and trigonometric functions from their properties, exactly in the spirit of this section.

The full guided solution is in Section 3.8:
https://beyondstep.keithmcnulty.org/ch03/addition-laws.html